STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-FIRST-ORDER-LOCALIZATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2599-2610 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 2612-2650, the complete author proof (37 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 11 of queue_examples-defos.json (pb=1758, stmt 2599-2610, proof 2612-2650).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00S1: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00S7: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00S0: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DZ4: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2383–2393. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DYS: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1230–1237. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A6A: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2650–2650. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let k be a field. Let B be a finite type k-algebra. Let $S \subset B$ be a multiplicative subset ideal such that if $\mathrm{Spec}(B) \to \mathrm{Spec}(k)$ is not smooth at $\mathfrak{q}$ then $S \cap \mathfrak{q} = \emptyset$. Let N be a finite B-module. Then there is a canonical bijection*

$$Exal_k(B, N) \to Exal_k(S^{-1}B, S^{-1}N)$$

Proof. This proof is the same as the proof of Lemma 0DZ4 but easier. We suggest the reader to skip the proof. The map is given by localization: given $0 \to N \to C \to B \to 0$ in $\mathrm{Exal}_k(B, N)$ we send it to the localization $S_C^{-1}C$ of C with respect to the inverse image $S_C \subset C$ of S. Compare with the proof of Lemma 0DYS.

The smooth locus of a morphism of schemes is open by definition. Let $J \subset B$ be an ideal cutting out the set of points in $\mathrm{Spec}(B)$ where $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is not smooth. Since $k \to B$ is of finite presentation the complex $NL_{B/k}$ can be represented by a complex $N^{-1} \to N^0$ where N^i is a finite B-module, see Algebra, Section 00S0 and in particular Algebra, Lemma 00S1. As B is Noetherian, this means that $NL_{B/k}$ is pseudo-coherent. For $g \in J$ the k-algebra B_g is smooth and hence $(NL_{B/k})_g = NL_{B_g/k}$ is quasi-isomorphic to a finite projective B-module sitting in degree 0. Thus $\mathrm{Ext}^i_B(NL_{B/k}, N)_g = 0$ for $i \geq 1$ and any B-module N. Finally, we have

$$\mathrm{Ext}^1_{S^{-1}B}(NL_{S^{-1}B/k}, S^{-1}N) = \mathrm{Ext}^1_B(NL_{B/k}, N) \otimes_B S^{-1}B = \mathrm{Ext}^1_B(NL_{B/k}, N)$$

The first equality by More on Algebra, Lemma 0A6A and Algebra, Lemma 00S7. The second because $\mathrm{Ext}^1_B(NL_{B/k}, N)$ is J-power torsion and elements of S act invertibly on J-power torsion modules. This concludes the proof by the description of $\mathrm{Exal}_A(B, N)$ as $\mathrm{Ext}^1_B(NL_{B/A}, N)$ given just above Lemma 0DZ4. □